EnergyMind Tunisia — Goal 4

Shared Learning Across the Noosphere:
a privacy-preserving, attack-aware collective defence for neighborhood nodes

1 Shared Learning Across the Noosphere

1.1 Why a single node is not enough

A neighborhood node sees only a few examples of each failure mode. In our simulated year, each falsification type occurs only 2–3 times *in the whole dataset*, that is at most once per node, and several types never occur in a given node at all (Table 1). The hardest cases for a lone detector (`selective_bypass`, `partial_reduction_bypass`, `meter_zeroing`, `gain_drift`, `clock_shift`, `negative_value`, `transformer_degradation`) are exactly the rare ones; the Goal 1 run on node N1 confirms that bypasses, biases and drifts are the cases a single node detects worst. A node that has never seen an attack cannot recognise it; a node that has will recognise it earlier if its neighbours warn it.

Table 1: Falsification events per node in the dataset. A 0 means the node never sees that attack in the year.

Falsification type	N1	N2	N3
`coordinated_fdi`	1	**0**	1
`fdi_evasion`	1	1	**0**
`fdi_inflate`	1	**0**	1
`fdi_replay`	1	**0**	1
`fdi_scale`	1	1	**0**
`meter_zeroing`	1	1	**0**
`partial_reduction_bypass`	1	1	1
`selective_bypass`	1	**0**	1

Goal 4 turns the local trust layer of Goals 1–3 into a *collective immune system*: nodes exchange what they *learned*, never what they *measured*.

The design rests on three principles:

P1. Share knowledge, not data. Raw readings, identifiers, explanations and logs never leave the node.

P2. Trust nothing shared by default. Every incoming artifact is treated like an unverified meter reading: it is validated, canary-tested locally, and adopted only with operator approval.

P3. Sharing can never trigger grid action. The escalation rules of Goal 3 stay in force whatever the shared model says; shared knowledge only changes *scores*, never *authority*.

1.2 Nodes and honest scope

The evaluation uses our simulated dataset (1,051,200 half-hourly rows, 60 meters, 7 feeders, seed 2025). Each node is one smart-metered neighborhood (Table 2); public Tunisian statistics were used only to calibrate the simulator, so no result here describes a real STEG grid. Node N1 is the

Phase 1 single-node prototype on which Goals 1–3 run; N2 and N3 are the other participants of the sharing design. As the challenge allows, multi-node sharing is a documented design in Phase 1: the protocol below is specified, and the experiments that would measure its benefit are planned in Section 1.8.

Table 2: Simulated nodes. Heterogeneity (climate, meter mix, feeder length, customer class) is what makes sharing useful; a common simulator limits how large it is.

Node	Profile	Meters	Feeders	Role in the experiments
N1	Coastal suburb	30	3	Largest node; local-baseline reference
N2	Hot interior town	15	2	Strong summer load, different climate regime
N3	Mixed residential/commercial	15	2	Different customer classes and load shapes

1.3 Architecture of the Noosphere

Figure 1 shows the architecture. Each node runs the full local trust pipeline and exposes only three narrow, versioned *channels* to a coordinator. A fourth, out-of-band *alert* path carries urgent attack warnings.

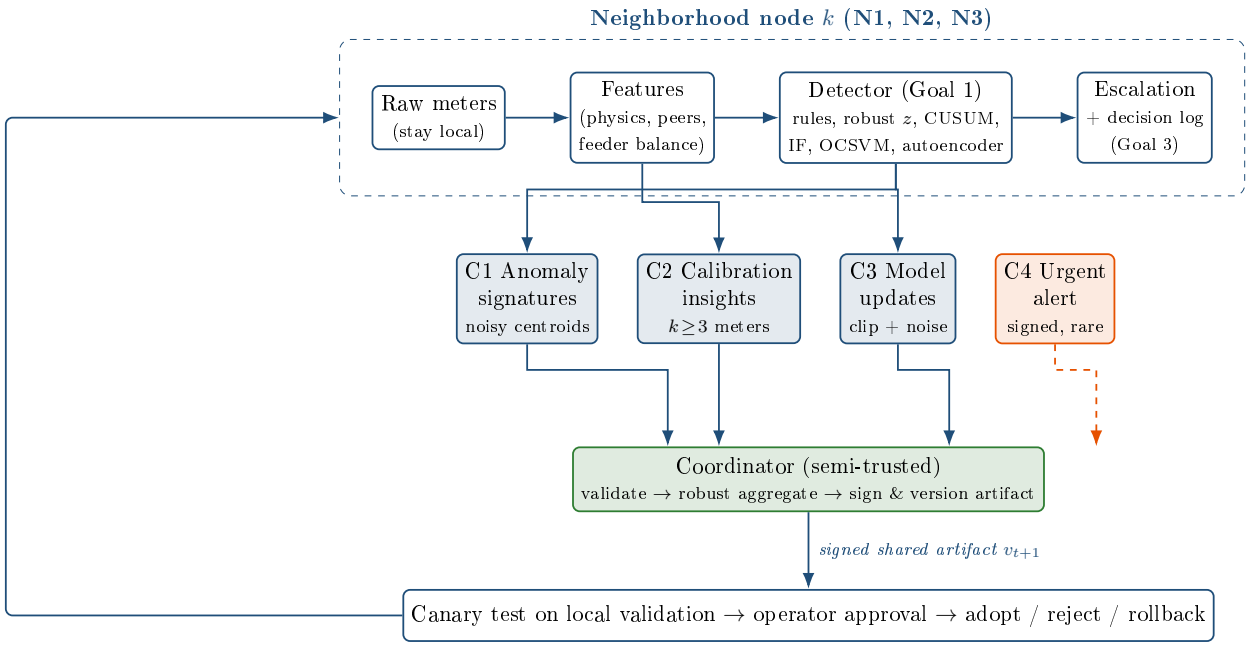


Figure 1: Noosphere architecture. Raw data and decisions stay inside the node; three protected channels and one urgent-alert path reach a coordinator, whose output is only adopted after a local canary test and human approval.

1.4 What is shared, what stays local, and why

Table 3: Sharing contract. Each row answers the challenge's three questions: what, what stays local, why.

Channel	Shared artifact	Safeguards	Why it helps
C1 Anomaly patterns	Per-node centroids of "trusted" and "untrusted" records in a robustly standardised feature space (Eq. 1); optional per-*family* centroids when $\geq n_{\min}$ events exist: physical bypass and data-only FDI (Goal 2 verdict `SUSPICIOUS_DATA`), sensor drift and corruption (`METER_FAULT`)	Feature clipping, Gaussian noise on released centroids, coordinate-wise median across nodes, count bucketing	Lets a node recognise an attack family it has seen too rarely to learn, without exchanging any time series
C2 Calibration insights	Median of $\log\big(E/(V\,I\,\mathrm{PF}\,\Delta t/1000)\big)$ per *meter model* on clean, non-exporting readings, with sample size bucket	Release a model only if ≥ 3 meters contribute; medians only; no meter IDs	Separates a model-wide calibration bias (firmware, batch) from a single faulty meter
C3 Model updates	Clipped, noised updates to a shared logistic-regression screening model	Update-norm clipping C, Gaussian noise σ, robust aggregation, secure aggregation (deployment)	Nodes with few rare events borrow strength from others; the shared model supports, never replaces, the local detectors of Goal 1
C4 Urgent alert	Signed "attack fingerprint": subtype family, feature-residual profile, affected meter-model, severity	Human-confirmed at the source node; rate-limited; expires; receivers treat it as a *prior*, not a verdict	Days-to-hours propagation of a confirmed coordinated attack

Stays local (never leaves the node): raw readings and timestamps; meter, customer and feeder identifiers; coordinates and dwelling data; per-meter reference values and the local median/MAD scaling statistics; per-record features, verdicts and SHAP explanations; operator notes and overrides; full decision logs; local detector models. *Reason:* half-hourly smart-meter traces reveal occupancy, sleep and appliance use [1, 2]; explanations and logs are accountability records that belong to the operator that made the decision.

1.5 Protocol

The protocol defines how independent neighborhood nodes prepare local information, share privacy-aware summaries or model updates, and evaluate the knowledge received from the coordinator. Raw meter readings and record-level decisions remain local. The protocol distinguishes four sharing channels: anomaly signatures (C1), calibration insights (C2), federated model updates (C3), and urgent alerts (C4). C1–C3 are regular knowledge-sharing mechanisms, whereas C4 is a separate, rare alert path.

1.5.1 Protocol overview

Figure 2 presents the complete protocol, from local data preparation to the local acceptance or rejection of a shared model. The multi-node communication and secure aggregation components

are documented design requirements; the figure describes the proposed architecture rather than claiming that the complete distributed system has been deployed.

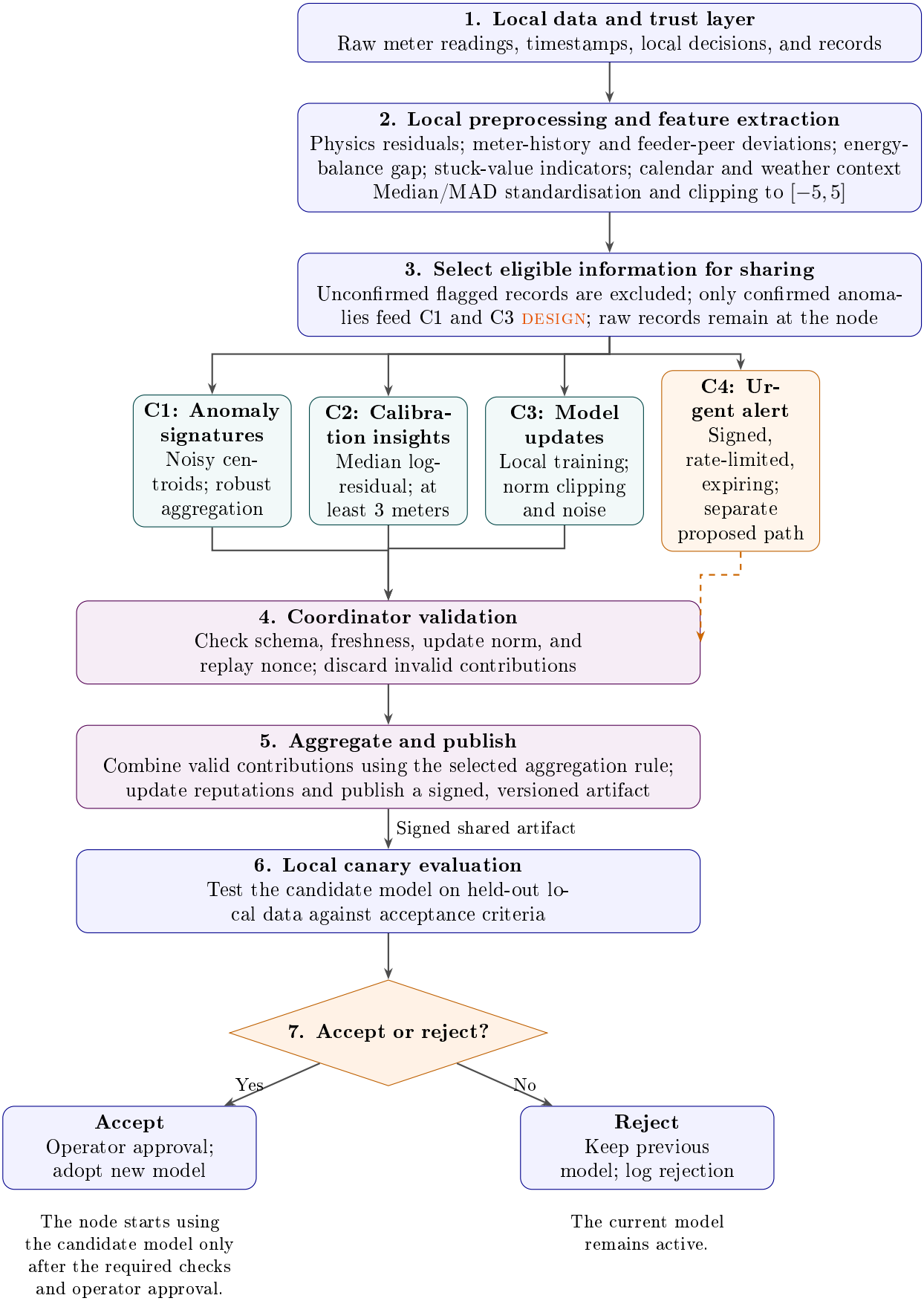


Figure 2: Proposed shared-learning protocol. Raw readings and record-level decisions remain local. C1–C3 carry regular shared information; C4 is a separate proposed urgent-alert path. A shared model is adopted only after local evaluation and operator approval.

1.5.2 Step 1: Local preprocessing and feature extraction

Each node processes its own meter data and constructs a feature vector describing possible anomalies. The features combine:

- physical consistency residuals between measured energy E and the estimate $VI\,\mathrm{PF}\,\Delta t$;
- deviations from the meter's own historical behaviour;
- deviations from peer meters on the same feeder;
- the feeder energy-balance gap;
- indicators of stuck or repeated values;
- calendar and weather context.

Since feature scales and distributions may differ between nodes, each node standardises feature j using its own median and median absolute deviation (MAD), and clips the result to a bounded interval:

$$z_{ij} = \mathrm{clip}\left(\frac{x_{ij} - \mathrm{med}_j}{1.4826\,\mathrm{MAD}_j}, -c, c\right), \qquad c = 5. \tag{1}$$

Here, x_{ij} is feature j for record i; med_j is the local median of feature j; MAD_j is its median absolute deviation; and $\mathrm{clip}(u, -c, c)$ restricts u to the interval $[-c, c]$. The factor 1.4826 is a conventional scaling factor for the MAD. The operation limits the influence of extreme feature values on subsequent shared statistics.

Two kinds of local records are kept apart DESIGN. Records that the local trust layer has flagged but no one has confirmed are excluded from everything that is shared, so an unverified anomaly can neither distort the shared picture of normal behaviour nor be presented as an attack. Records whose cause is confirmed (by an operator, or in the prototype by the dataset labels `gt_*`) are the *only* anomalies used for the "untrusted" and family centroids of C1 and for the screening model of C3. This is a documented design rule and should not be interpreted as proof that the complete multi-node exclusion mechanism has been deployed.

1.5.3 Step 2: Prepare the information to be shared

After local preprocessing, a node prepares the relevant artifact for the selected sharing channel. The four channels serve different purposes and must not be treated as interchangeable.

C1: Anomaly signatures. C1 shares compact representations of anomaly patterns, using centroids of trusted and untrusted records in the standardised feature space. When enough examples are available, centroids may also be maintained for anomaly families such as physical bypass, data-only false data injection (FDI), and sensor drift.

Before release, centroids use feature clipping and Gaussian noise; the coordinator combines the received summaries using coordinate-wise medians. Count bucketing is also specified as a safeguard. This allows nodes to learn from patterns observed elsewhere without exchanging individual meter time series.

C2: Calibration insights. C2 shares a robust summary of the calibration behaviour of a meter model. For node k and meter model m, the summary is released only when at least three meters of that model are available locally, $n_{k,m} \geq 3$:

$$\widetilde{r}_{k,m} = \mathrm{med}_{i \in m}\,\mathrm{med}_t \log\left(\frac{E_{it}}{V_{it} I_{it}\,\mathrm{PF}_{it}\,\Delta t/1000}\right). \tag{2}$$

Here, E_{it} is the measured energy for meter i at time t; V_{it} is voltage; I_{it} is current; PF_{it} is power factor; and Δt is the measurement interval (0.5 h; the factor 1000 converts W·h to kWh). Only clean readings with $V > 0$, no PV export and $E > 0$ are used: while a home exports, its imported

energy is 0 and the logarithm is undefined. The inner median summarises the time-dependent residual for each meter, while the outer median summarises the meters belonging to model m.

The coordinator then takes the median of the summaries received from eligible nodes:

$$\widetilde{r}_m = \mathrm{med}_k\, \widetilde{r}_{k,m}. \tag{3}$$

A value consistently far from zero across several nodes may indicate a systematic bias associated with that meter model. A deviation observed for only one meter is more consistent with an individual meter fault. This is a diagnostic indication, not by itself proof of the cause.

On our dataset C2 is a negative control. The three meter models differ only by communication technology (GPRS, NB-IoT, PLC), all meters have accuracy class 1.0, and no model-wide bias was injected. Computed on the clean data, $\widetilde{r}_{k,m}$ lies within ± 0.0002 (i.e. $\pm 0.02\%$) for every node and model, so any larger value would be a false alarm. With the $n_{k,m} \geq 3$ rule, the PLC model (5 meters in N1, none in N2, 2 in N3) is released by N1 only, so its "median across nodes" rests on a single node. A positive test is planned (Section 1.8): a +3% gain is injected on all NB-IoT meters *inside the experiment code*, leaving the dataset unchanged; C2 must report it, while a single biased meter (`bias_step`) must not move the model-level value.

C3: Federated model updates. C3 allows nodes to contribute to a shared screening model without transferring their local training records. At round t, node k trains the current model w_t on its local dataset $\mathcal{D}_k$. The resulting update is

$$\Delta_k = \mathrm{LocalTrain}(w_t; \mathcal{D}_k) - w_t. \tag{4}$$

The update Δ_k represents the change produced by local training. It is not the raw dataset and does not need to contain individual meter readings.

C4: Urgent alerts. C4 is a separate proposed path for rare, urgent information. An alert may contain a signed attack fingerprint describing the subtype family, feature-residual profile, affected meter model, and severity. The design requires confirmation by a human operator at the source, rate limiting, and expiration. Receiving nodes should treat the alert as a prior indication to investigate, not as a definitive verdict. The full distributed alert path remains a design requirement.

The dataset contains a ready-made case. A `coordinated_fdi` attack (three meters of one feeder falsified together) hits N3 from 1 to 28 July 2025 and N1 from 14 October 2025. Once N3's operator confirms the July attack, its fingerprint reaches N1 three months before N1's own attack. Because Goal 1's physics rule already flags data-only injection in N1 from the first day, the expected benefit of the alert is not earlier detection but a faster and more confident diagnosis: the receiving node can name a confirmed attack family instead of an unexplained anomaly.

1.5.4 Step 3: Protect model updates before sharing

Before a C3 update is shared, the node clips its Euclidean norm and adds Gaussian noise:

$$\widetilde{\Delta}_k = \Delta_k \min\left(1, \frac{C}{\|\Delta_k\|_2}\right) + \mathcal{N}\left(0, \frac{\sigma^2 C^2}{K} I\right). \tag{5}$$

The symbols have the following meanings:

- C is the maximum allowed update norm;
- $\|\Delta_k\|_2$ is the Euclidean norm of the local update;
- σ is the noise multiplier;
- K is the number of participating nodes;
- I is the identity matrix;

- $\mathcal{N}(0, \frac{\sigma^2 C^2}{K} I)$ is the zero-mean Gaussian noise distribution used by the design.

If the update norm is no greater than C, the clipping factor is one. If it exceeds C, the factor reduces the update until its norm is C. Gaussian noise is then added as a privacy safeguard.

Clipping limits the influence of a single large update, while noise is intended to reduce the information revealed by the shared update. However, clipping and adding noise alone do not establish a formal differential-privacy guarantee. Such a guarantee would require a specified privacy analysis and accounting procedure.

1.5.5 Step 4: Validate and aggregate contributions

The coordinator first checks received contributions against the announced protocol version and feature schema. It also checks freshness, update norms, and replay nonces, discarding invalid contributions. The protocol requires authenticated and encrypted communication; secure aggregation remains a deployment design requirement.

For valid C3 updates, the coordinator applies the selected aggregation rule and updates the shared model:

$$w_{t+1} = w_t + \mathrm{Agg}\Big(\widetilde{\Delta}_1, \ldots, \widetilde{\Delta}_K\Big). \tag{6}$$

Here, Agg is the aggregation function, and w_{t+1} is the candidate shared model for the next round. Three aggregation rules are considered.

FedAvg. FedAvg takes the arithmetic mean of the protected updates:

$$\mathrm{Agg}_{\mathrm{FedAvg}} = \frac{1}{K} \sum_{k=1}^{K} \widetilde{\Delta}_k. \tag{7}$$

This provides a simple way to combine the contributions of all participating nodes. However, an unusually large or malicious update may still affect the average, which motivates robust alternatives.

Coordinate-wise median. The coordinator calculates the median separately for each model coordinate j:

$$(\mathrm{Agg}_{\mathrm{med}})_j = \mathrm{med}_k \left(\widetilde{\Delta}_{k,j}\right). \tag{8}$$

Compared with the mean, the coordinate-wise median is less sensitive to extreme values in individual coordinates. Its robustness has limits, especially when few nodes participate or several nodes collude.

Reputation-weighted aggregation. The coordinator can also weight updates according to node reputation and how far each update deviates from the median update:

$$\mathrm{Agg}_{\mathrm{rep}} = \sum_{k=1}^{K} \rho_k \widetilde{\Delta}_k, \qquad \rho_k \propto r_k \exp\left(-\frac{d_k}{\tau}\right), \tag{9}$$

where

$$d_k = 1 - \cos\Big(\widetilde{\Delta}_k,\, \mathrm{med}_{k'}\, \widetilde{\Delta}_{k'}\Big). \tag{10}$$

In this expression, $r_k \in [0, 1]$ is the reputation of node k, d_k measures the cosine distance between its update and the coordinate-wise median update, and τ controls the effect of that distance on the weight. The proportional weights are normalised so that they sum to one. Reputation is designed to decrease after rejected rounds and recover after accepted rounds.

The three aggregation rules are alternatives; they are not three successive operations that must all be applied to the same update. The choice of aggregation rule should be stated for each experiment or deployment configuration.

For C1 anomaly centroids, the coordinator instead applies coordinate-wise medians to the protected summaries. C2 calibration summaries are combined using the median across eligible nodes, as defined in Equation 3.

1.5.6 Step 5: Publish the shared artifact

After validation and aggregation, the coordinator publishes a signed, versioned artifact containing the relevant shared result. The coordinator announces the round identifier, protocol version, feature-schema identifier, clipping and noise parameters (c, C, σ), and deadline. Nodes sign and submit their contributions through the specified authenticated, encrypted channel.

Versioning allows nodes to identify which shared artifact they have received. Signatures support integrity and origin verification, while the validation checks help reject stale or malformed contributions. These protections describe the intended protocol; they do not imply that a complete secure multi-node transport and aggregation system has been implemented.

1.5.7 Step 6: Local canary evaluation and adoption

A node does not automatically adopt a newly published model. It first evaluates the candidate on held-out local canary data and checks the following criteria:

- ROC-AUC must not decrease beyond an allowed margin Δ_{AUC};
- the false-positive rate must remain within the defined alarm budget;
- genuine `grid_event` and `equipment_failure` records must not be misclassified as data-quality problems;
- recall per attack *family* (physical bypass, data-only FDI, sensor drift, corruption) must not decrease. Per-type recall cannot serve as a gate: each falsification type occurs at most once per node, so its recall can only be 0 or 1. A family gates adoption only where the node holds at least three events of it (Table 4); otherwise its result is reported to the operator without blocking adoption.

Table 4: Events per attack family and node, and whether the family can gate adoption (≥ 3 events).

Family	N1	N2	N3
Physical bypass	3 (gate)	2	2
Data-only FDI	5 (gate)	2	3 (gate)
Sensor drift	8 (gate)	6 (gate)	4 (gate)
Corruption	25 (gate)	14 (gate)	14 (gate)

The numerical thresholds for these criteria must be specified before deployment. If the candidate passes the checks and receives the required operator approval, the node adopts it. Otherwise, the node retains its previous model and logs the rejection.

This local evaluation step limits the risk of adopting a shared model that is less suitable for a particular neighborhood. It does not, by itself, prove that the model is safe against every attack or that the protocol improves detection performance.

1.5.8 Implementation status and limitations

The protocol combines local processing, protected information sharing, robust aggregation, and local model evaluation. Its claims must nevertheless be interpreted in light of the current implementation scope:

- The experimental data are synthetic; they are not real STEG meter readings.
- The complete multi-node sharing experiment has not been run. The current work does not establish a measured performance improvement from cross-node collaboration.
- Network transport, node authentication, and secure aggregation remain documented design requirements.

- The noise mechanisms described here should not be presented as a formal differential-privacy guarantee without a complete privacy analysis.
- Robust aggregation has limitations with a small number of participants. With only three nodes, for example, a coordinate-wise median cannot guarantee protection against coordinated malicious contributions.

Accordingly, this section specifies the proposed shared-learning protocol and its safeguards. What exists as code today are the protocol primitives in `src/energymind/shared/`: update clipping and Gaussian noise (Eq. 5), FedAvg and coordinate-wise median aggregation, and the C2 calibration summary. The multi-node rounds, the coordinator, transport security and secure aggregation are design.

1.6 Privacy: what is protected and what is not

Layered protections, each with a distinct role: locality (no raw data), clipping (bounded influence), Gaussian noise (plausible deniability for individual contributions [9, 10]), group suppression ($k \geq 3$, in the spirit of k-anonymity [3], *not* a DP guarantee) and secure aggregation so the coordinator sees only the sum [8]. Gradient-style updates can leak training data if sent in the clear [12], which is why secure aggregation is part of the deployment design.

For FedAvg with all K nodes participating in each of R rounds, Gaussian noise multiplier σ and clipping bound C, the Rényi-DP accounting of [11] gives

$$\varepsilon(\delta) = \min_{\alpha>1} \left[\frac{R\alpha}{2\sigma^2} + \frac{\ln(1/\delta)}{\alpha - 1} \right] = \frac{R}{2\sigma^2} + \sqrt{\frac{2R\ln(1/\delta)}{\sigma^2}}, \qquad \delta = 10^{-5}. \tag{11}$$

This assumes the summed noise (each node adds $\sigma^2 C^2/K$) is observed only after secure aggregation. Table 5 evaluates Eq. (11); the values are computed from the formula, not measured on our data.

Table 5: ε at $\delta = 10^{-5}$ from Eq. (11) (node-level, FedAvg only).

σ	$R = 10$	$R = 26$	$R = 52$
1	20.2	37.5	60.6
2	8.8	15.5	23.8
4	4.1	6.9	10.3
8	2.0	3.3	4.7

Honest reading. Even a weekly schedule ($R = 52$) needs $\sigma \approx 8$ for $\varepsilon < 5$, and with $K = 3$ that much noise will likely cost accuracy, which is exactly the privacy–utility trade-off the planned experiment will measure (Section 1.8); no utility number is claimed in Phase 1. Eq. (11) covers FedAvg only; it does not cover signatures, calibration summaries, class counts, participation metadata, or the median and reputation rules, which need their own analysis. Until secure aggregation and transport security are implemented and the accounting is reviewed, the system is *privacy-aware*, not *provably private*.

1.7 Threat model and collective defence

Table 6: Threat–defence matrix and what remains uncovered.

Threat	Example	Defence	Residual risk
Honest-but-curious coordinator	Inspects each node's update/summary	Secure aggregation, noise, bucketed counts	Not implemented (design)
Network eavesdropper / tamperer	Reads or alters artifacts in transit	TLS, signed and versioned artifacts	Key management out of scope
Curious peer node	Infers another node's behaviour from the shared model	Clipping, noise, $k \geq 3$ suppression	No formal bound beyond Eq. (11)
Poisoned update	Sign-flipped or oversized Δ	Norm clip, median / reputation, canary test	Tolerates at most 1 of 3 arbitrary outliers
Replay / stale update	Resends an old Δ	Round nonce, freshness check	—
Backdoor / adaptive attack	Update crafted to blind the model to one subtype [13]	Canary per-family recall, human approval, rollback	Not proven robust; basic defences are known to be weak
False alert injection (C4)	Fake "attack fingerprint" to cause alarms or desensitise	Source-node human confirmation, signature, rate limit, expiry, prior-only use	Sybil nodes need controlled registration
Collusion / Sybil	Majority of nodes (or fake identities) collude	Controlled node registration, governance rules	**Not defended**: median fails if ≥ 2 of 3 collude

Over-trust and under-trust of shared knowledge. The Goal 3 concern about operators applies to the network as well. *Over-trust* (blindly adopting a shared model) is prevented by the canary test, per-family recall checks, human approval and rollback. *Under-trust* (rejecting useful updates forever) is monitored: every rejection is logged with its reason and the shadow-model comparison, so a persistent pattern of rejected-but-beneficial updates is visible to reviewers. Shared outputs only adjust anomaly scores; they do not change the autonomy thresholds or trigger any action on the grid, and Goal 3's non-tunable routing rules (R0–R3) apply whatever is shared.

1.8 Evaluation plan

No experiment across nodes has been run yet, and no performance number for sharing is claimed. The planned evaluation, all on the existing dataset (no change to the data), is:

1. **Splits.** A calendar split does not work: in N1, November–December contain no falsification and no sensor-bias event. Models are therefore evaluated by cross-validation with folds grouped by event (`gt_event_id`), as in Goals 1–2; results are reported per event, with intervals. The `gt_*` columns are never model inputs.
2. **Transfer to unseen attacks.** For every attack type that is missing in some node (Table 1), train on the nodes that have it and score its events while the receiving node's own data contain no example of it; compare local only, local + C1 centroids, and local + C3 (FedAvg vs. coordinate-wise median). Measure per-family recall at a fixed false-positive rate.
3. **Privacy–utility.** Repeat the C3 run for $\sigma \in \{1, 2, 4, 8\}$ and plot detection against the ε of Table 5.
4. **Corrupted node.** One node sends sign-flipped or scaled updates; compare FedAvg, median and reputation-weighted aggregation, and check that the canary rejects the poisoned artifact.

5. **C2.** Negative control on the clean data (all models ≈ 0), then the +3% NB-IoT injection done in code.
6. **C4.** N3's July coordinated attack as a prior for N1 before 14 October: diagnosis time and confidence with and without the alert.

1.9 Limitations

1. All data and anomalies are synthetic; results may reflect the simulator and cannot establish real-grid performance. External validation (e.g. Low Carbon London, UCI) is needed.
2. The three nodes come from one generator, so their heterogeneity understates that of independent real operators.
3. With $K = 3$, robust aggregation has almost no redundancy and strong DP is costly (Table 5).
4. The planned corrupted-node test (Section 1.8) covers sign-flipped and scaled updates only, not backdoors, collusion, adaptive attacks, Sybil identities or a compromised coordinator.
5. The simulated evasion attack is a random-walk scaling, not the gradient-based attacks of the literature.
6. Each falsification type occurs 2–3 times in the dataset and at most once per node, so per-type results are case studies; families are used for decisions, with wide uncertainty.

References

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